Roll No. | | | | | | | | | | |

Total No. of Pages : 03

Total No. of Questions : 13

B.Pharmacy (Sem.-8)

# BIOSTATISTICS AND RESEARCH METHODOLOGY

Subject Code : BP801T

M.Code : 79764

Date of Examination : 21-05-2024

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

## SECTION-A

1. Write Briefly :

(a) What are the ideal characteristics of measure of dispersion?

(b) What are different types of frequency distribution?

(c) Distinguish between Binomial and Poisson distribution.

(d) Define Null Hypothesis and Alternate Hypothesis.

(e) What is the need for research?

(f) Give major applications of minitab statistical software in clinical trials.

(g) Differentiate between observational and experimental studies.

(h) Define Factorial Designs.

(i) Enlist practical components of clinical trial problems.

(j) Differentiate between Correlation and Regression analysis.

Chandigarh ... LIBR ... Landran



(S29)-989

## SECTION-B

2. Test following data for statistical significance. If Significant difference exists among groups, then apply Fisher's Least Significant Difference test for multiple comparison.

| A | B | C | D |
|---|---|---|---|
| 14 | 22 | 58 | 34 |
| 18 | 24 | 54 | 33 |
| 17 | 28 | 59 | 38 |
| 15 | 29 | 53 | 35 |
| 12 | 20 | 56 | 37 |
| 18 | 28 | 57 | 42 |
| 19 | 29 | 59 | 39 |
| 11 | 24 | 57 | 35 |
| 10 | 23 | 53 | 33 |
| 17 | 21 | 55 | 31 |

3. (a) Two automatic filing machines A and B are used to fill tea in 500 gm cartons. A random sample of 100 cartons on each machine showed the following :

| Tea content (in gms.) | A | B |
|---|---|---|
| 470-480 | 12 | 10 |
| 480-490 | 18 | 15 |
| 490-500 | 20 | 24 |
| 500-510 | 22 | 20 |
| 510-520 | 24 | 18 |
| 520-530 | 4 | 13 |

Comment on the performance of two machines,

(b) Write a note on 'Karl Pearson's Coefficient of Correlation'.



June-2024

4. **Write notes on :**

   (a) Wicoxon Rank Sum test

   (b) Plagiarism

## SECTION-C

5. Write a note on optimization Techniques.

6. Discuss advantages of factorial designs in analysis of experiments?

7. Explain Confounding system for two level factorials.

8. Compute mean, mode and median from the given table for the heights of the students with the corresponding frequencies :

| Heights | 57-58 | 58-59 | 59-60 | 60-61 | 61-62 | 62-63 | 63-64 | 64-65 | 65-66 | 66-67 | 67-68 | 6 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Frequency | 10 | 12 | 14 | 20 | 22 | 43 | 37 | 14 | 8 | 12 | 5 | |

9. Explain curve fitting by method of least squares.

10. Describe various phases of Clinical trials.

11. What are different types of sampling? Explain probability sampling methods.

12. What are different types of graphs for representation of statistical data? Explain respo surface plot and counter plot graph.

13. Write a detailed note on R-Online statistical software.

NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on a paper of Answer Sheet will lead to UMC against the Student.